

### Scalability & Future Plan

|                      |                          |
|----------------------|--------------------------|
| <b>Date</b>          | 15 March 2024            |
| <b>Team ID</b>       | 6aba25588343c5cff063937b |
| <b>Project Name</b>  | Comic craft AI           |
| <b>Maximum Marks</b> | 1 Mark                   |

#### Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

#### Current System Limitations:

| S.No | Limitation                                             | Impact                                                                                | Priority to Address (High / Medium / Low) |
|------|--------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------|
| 1    | Dependence on external AI API                          | Story generation may be affected if the AI service is unavailable or slow             | High                                      |
| 2    | Limited handling of high numbers of concurrent users   | Response time may increase when many users generate stories simultaneously            | High                                      |
| 3    | Limited persistent storage for generated comic content | Large amounts of generated stories and character data may become difficult to manage. | Medium                                    |

#### Scalability Plan:

| S.No | Scalability Aspect | Current State                          | Proposed Upgrade / Solution                                                                      |
|------|--------------------|----------------------------------------|--------------------------------------------------------------------------------------------------|
| 1    | User Load          | Suitable for a limited number of users | Deploy on scalable cloud infrastructure and use load balancing to support more concurrent users. |
| 2    | Data Storage       | Limited/local storage                  | Integrate a scalable database and cloud storage for stories, characters, and generated content.  |
| 3    | Performance        | Suitable for normal usage              | Introduce caching, asynchronous processing, and optimized API requests to improve response time. |

|   |          |                                             |                                                                                                  |
|---|----------|---------------------------------------------|--------------------------------------------------------------------------------------------------|
| 4 | Security | Basic API-key and environment configuration | Implement authentication, authorization, secure API-key management, input validation, and HTTPS. |
|---|----------|---------------------------------------------|--------------------------------------------------------------------------------------------------|

Future Roadmap:

| Phase   | Planned Feature / Enhancement                               | Target Timeline | Expected Impact                                                                           |
|---------|-------------------------------------------------------------|-----------------|-------------------------------------------------------------------------------------------|
| Phase 2 | User accounts and saved comic stories                       | 1–2 Months      | Users can securely save and manage their generated comics.                                |
| Phase 3 | AI-generated comic images and character illustrations       | 3–4 Month       | Makes the project more visual and improves the comic-creation experience.                 |
| Phase 4 | Cloud deployment, scalable database, and multi-user support | 5–6 Months      | Allows Comic Craft AI to support a larger number of users and handle increased workloads. |